INCH-POUND

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PERFORMANCE SPECIFICATION

CLOTH, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of waterproof cloth.

1.2 Classification. The following types of cloth cited in this specification are as follows (see 6.2).

Type I - Camouflage Green 483
Type II - Deleted
Type III - Woodland Camouflage
Type IV - Desert Camouflage

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions, clarifications) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FNS, 2800 South 20th Street, Philadelphia, PA 19145-5099 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter..

AMSC N/A **FSC 8305**

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-961 - Department of Defense Standard Practice For Defense Specifications
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Unless otherwise indicated. copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk. 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS

(Applications for copies of referenced documents should be addressed to U. S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328.)

FEDERAL TRADE COMMISSION

Rules and Regulations Under-the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

U. S. ARMY CENTER FOR HEALTH PROMOTION AND PREVENTATIVE MEDICINE

Contact Allergy: Predictive Testing in Humans", Advances in Modern Toxicology

(Applications for copies of referenced documents should be addressed to U. S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg. E-2100, Aberdeen Proving Grounds, MD 21010-5442.)

FEDERAL ACQUISITION REGULATIONS (FAR)

FAR 52.209 - First Article Testing

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(Applications for copies of referenced documents should be addressed to the Superintendent of Documents, P. O. Box 371954, Pittsburgh, PA 15250-7954.)

BUREAU OF ALCOHOL,TOBACCO AND FIREARMS,
DEPARTMENT OF THE TREASURY

Formulas for Denatured Alcohol (27 CFR Part 21)

(Applications for copies of referenced documents should be addressed to the Bureau Of Alcohol, Tobacco And Firearms, Department Of The Treasury, 1200 Pennsylvania Ave. Washington, DC 20226.)

ENVIRONMENTAL PROTECTION AGENCY

Regulations for the Enforcement of the Federal Insecticide, Fungicide and Rodenticide Act (40 CFR Part 162)

(Applications for copies of referenced documents should be addressed to the Environmental Protection Agency, 401 M Street SW, Washington, DC 20460.)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT & ENGINEERING CENTER

2-1-1516B Pattern, Woodland
2-1-2240 Desert Camouflage pattern, Three Color

(Copies of drawings are available from the U. S. Army Natick Research, Development and Engineering Center, ATTN: SSCNC-UX, Natick, MA 01760-5017.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-751 - Water Resistance of Cloth; Water Permeability; Hydrostatic Pressure Method
ASTM-D-1424 - Tear Resistance of Woven Fabrics by Falling Pendulum (Elmendorf) Apparatus
ASTM-D-1776 - Standard Practice for Conditioning Textiles For Testing
ASTM-D-3776 - Mass Per Unit Area (Weight) of Woven Fabric
ASTM-D-3886 - Abrasion Resistance of Textile Fabrics (Inflated Diaphragm Method)
ASTM-D-5035 - Breaking Force And Elongation of Textile Fabrics (Strip Force)

(Applications for copies of referenced documents should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19426-2959.)

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TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC-8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC-111 - Weather Resistance: Exposure to Natural Light And Weather Through Glass
AATCC-169 - Weather Resistance of Textiles: Xenon Lamp Exposure

(Applications for copies of referenced documents should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI Method T-451, Stiffness, Preferred Procedure (1)

(Applications for copies of referenced documents should be addressed to TAPPI Press, Technology Park/Atlanta, P.O. Box 105113, Atlanta, GA 30348-5113.)

SEARS ROEBUCK AND COMPANY FABRIC DEFECT SCALES

(Fabric defect replica kits are available from Sears Roebuck and Company, Department 817, (ATTN: BSC 23-29), Sears Tower, Chicago, IL 60684).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.2.

3.2 Standard sample. The waterproof cloth shall match the standard sample for shade and appearance and shall be equal to or better than the sample for colorfastness (see 6.4).

3.3 Performance requirements. The types I, III and IV cloth shall conform to the requirements specified in Table I and 3.4 through 3.8.

TABLE I. Requirements

Characteristic	Requirements
Weight Oz/yd^2 (max)	3.25
Breaking Strength, lbs. (min):	
Warp	50
Filling	50
Tearing Strength, grams (min):	
Warp	700
Filling	900

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TABLE I. Requirements

Characteristic	Requirements
Hydrostatic Resistance psi (min):	
Initial	60
After accelerated weathering	25 1/
After abrasion	35
After strength of coating	35
After high humidity	40 1/
Colorfastness (min)	
To Accelerated Weathering	
Type I	No appreciable change 2/
Types III & IV	5/
To high humidity	
Type I	No appreciable change 2/
Types III & IV	5/
Stiffness, mm: (max):	
At standard conditions	70
At 0°F	100
Adhesion of coating, lbs/2 inch width (min)	7.0 6/
Blocking, scale rating (max)	No. 1
Water wicking, inches (max)	1/8
Resistance to Leakage (min.)	No leakage 3/
Water absorption, % (max)	7.5
Resistance to insect repellent (min)	4/

1/ The cloth shall not become stiff and brittle nor soft and tacky and there shall be no evidence of cracking or crazing under visual examination.
2/ "Appreciable change in color" means a change that is immediately noticeable on comparison of the test specimen with the original, unexposed sample. If closer inspection or a change of angle of light is required to make apparent a slight change of color, the change is not considered appreciable.
3/ Leakage is defined as the appearance of water anywhere within the 4 1/2 inch diameter test area.
4/ The cloth shall show no lifting, no tackiness, no solution, no pickoff, no adherence to itself greater than scale rating (2) (slight blocking) (see 6.5.1).
5/ Equal to or better than the standard sample.
6/ Applicable only if a film or coating is used.

3.4 Color. (types defined in accordance with FED-STD-595).

3.4.1 Type I. The color of the cloth shall be Camouflage Green 483 matching color chip 34094 of FED-STD-595 (see 3.2 and 6.4).

3.4.2 Type III. The color of the cloth shall be Light Green 354, Dark Green 355, Dark Brown 356 and Black 357 with each area matching the specific colors of the Woodland Camouflage pattern on the standard sample (see 3.2 and 6.4).

3.4.3 Type IV. The color of the cloth shall match shades Light Tan 492, Light Brown 493 and Light Khaki 494. Each area shall match the specific colors of the pattern or the standard sample (see 3.2 and 6.4).

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3.4.3.1 Colorfastness to crocking (all types). All colors (both sides) except Black 357 shall show an AATCC Chromatic Transference Scale rating for crocking not less than 3.5. Black 357 shall show an AATCC Chromatic Transference Scale rating for crocking not less than 1.5 when tested as specified in 4.3.2.

3.5 Spectral reflectance. The spectral reflectance requirements of the Type I Camouflage Green 483 cloth and for Type III, Woodland Camouflage cloth shall conform to the requirements specified in Table IIa. The spectral reflectance requirements of the Type IV Desert Camouflage cloth shall conform to the requirements specified in Table IIb.

TABLE IIa. Spectral reflectance requirements; Reflectance (%)

	Type I		Type III				
Wavelengths	Camouflage Green 483		Light Green 354		Dark Green 355 & Brown 356		Black 357
Nanometers	Min	Max	Min	Max	Min	Max	Max
600	5	12	8	20	3	13	10
620	5	12	8	20	3	13	10
640	5	12	8	20	3	13	10
660	5	13	8	22	3	13	10
680	6	15	8	36	3	22	10
700	9	21	14	60	8	46	10
720	15	30	26	78	20	66	10
740	24	42	40	90	30	80	10
760	32	50	50	92	32	88	10
780	38	56	55	92	32	90	10
800	41	60	55	92	32	91	10
820	43	63	55	92	32	92	10
840	45	65	55	92	32	92	10
860	46	66	55	92	32	92	10

TABLE IIb. Spectral reflectance requirements; Reflectance (%)

	Type IV					
Wavelengths	Light Tan 492		Light Brown 493		Light Khaki 494	
Nanometers	Min	Max	Min	Max	Min	Max
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46
760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58

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3.6 Pattern execution, type III and type IV. The pattern shall reproduce the design as depicted on Drawing 2-1-1516B (type III) and 2-1-2240 (type IV) for design registration, color placement and pattern repeat. The pattern repeat for the type III cloth shall be 27.25 +2.00 -2.50 inches in the warp direction. The pattern repeat for the type IV cloth shall be 16.75 +1.25 -1.75 inches in the warp direction (see 4.4.13).

3.7 Toxicity. The cloth shall not be toxic to the skin or eyes when used as intended (see 4.4.14).

3.8 Mildew resistance. The cloth shall be mildew resistant. If plastizers are used they should be restricted to phosphate or pthalate esters for the purpose of providing mildew resistance. A certificate of compliance will be acceptable for this requirement (see 4.4.15).

3.9 Face side identification. The face side of the waterproof cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" on each end of each individual piece.

3.10 Fiber identification. When applicable, each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.11 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects listed in 4.3.1 and shall be tested for the characteristics specified in Table IV.

4.3 Conformance inspection. Sampling for inspection shall be as specified in the contract or purchase order.

4.3.1 End item examination.

4.3.1.1 Yard by yard examination. During the yard by yard examination, each roll in the sample shall be examined for the effects specified in Table III. The yardage of each roll in the sample shall be examined on both sides. The yardage shall be given a through-light inspection for pinholes and thin areas. The through-light inspection shall be performed in accordance with MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The sample unit shall be one linear yard.

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Table III End item visual defects

Defect	Classification	
	Major	Minor
Any hole, cut, tear or scratch, including edges	101	
Abrasion resulting in a thin or weak place	102	
Multiple floats or skips, 1/2 inch or more in either warp or filling direction of face fabric	103	
Any pit, blister, tunnel, bubbles, or separation of finish (if finished)	104	
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure, adhesion of surfaces against each other, or any diagonal distortion of face side surface	105	
Any solid lump, defined as a slub C or 4, or knot which exceeds level C on the respective Sears Fabric Defect Scale (see 2.2)	106	
Fabric edges rolled, folded, doubled, scalloped or wavy	107	
Any spot, stain 1/ or foreign matter 2/	108	
Any odor other than that which is characteristic of the component materials of the cloth		201
Any tackiness	109	
Any pinhole	110	
Any area without waterproof finish (i.e., coating or laminating film)	111	
Any scorch or burn	112	
Multiple floats or skips 1/2 inch or more in either warp or filling direction of face fabric	113	
Not clean	114	
Camouflage pattern (face side)		
Any skitteriness, feathering or spew of pattern exceeding that shown by standard sample (Types III and IV only applies to face side)	115	
Pattern design or color placement not equal to pattern drawing (Types III and IV only; applies to face side)	116	
Pattern colors not showing sharp demarcations between adjacent colors (Types III and IV only; applies to face side)	117	
Pattern colors are not properly registered (Types III and IV only; applies to face side)	118	
Warpwise pattern repeat less than 24.75 inches or more than 28.50 inches (Type III only; applies to face side)	119	
Warpwise pattern repeat less than 15.50 inches or more than 18.50 inches (Type III only; applies to face side)	120	

1/ Clearly visible at normal inspection distance (approximately 3 feet).
2/ For the back side, any spot, stain, off-shade area, or discoloration that is a result of the distortion of a backing fabric (if used) or a result of uneven dyeing of a backing fabric shall not be scored for this condition. Foreign matter shall be defined as waste, fly, or extraneous material that has been formed into the fabric system.

4.3.1.2 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for proper identification. Any roll in the sample that is not labeled or ticketed in accordance with the Textile Fiber Products Identification Act or any piece not marked with the word "Face", when required, shall be considered a roll identification defect.

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4.3.1.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined on the face side for shade and appearance. If any roll in the sample is off shade or shaded side to side, side to center, or end to end, or if any roll does not have the same appearance as the standard sample, it shall be considered a shade and appearance defect.

4.3.2 End item testing. The cloth shall be tested for the characteristics listed in Table IV. The methods of testing specified wherever applicable and as listed in Table III shall be followed. The sample unit for testing shall be 3 continuous yards full width of the finished cloth. The lot shall be unacceptable if any sample unit fails to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests

Characteristic	Test Method*
Weight	ASTM D 3776 Method C (small swatch of fabric method)
Breaking strength	ASTM D 5035 (1C-E or 1C-T)
Tearing strength	ASTM D 1424
Colorfastness: To accelerated weathering To high humidity	 AATCC-111A, Opt.4 1/ 5512 4.4.2
Hydrostatic Resistance Initial After accelerated weathering After abrasion After strength of coating After high humidity	 4.4.1 AATCC-111A, Opt.4 1/ or 4.4.8 & 4.4.1 4.4.6 & 4.4.1 5972 2/ & 4.4.1 4.4.2 and 4.4.1
Water wicking	4.4.3
Resistance to leakage	4.4.4

* Paragraph numbers alone indicate the paragraph numbers in section 4 of this document.

TABLE IV. End item tests

Characteristic	Test Method *
Resistance to diethyltoluamide	4.4.5
Abrasion resistance	4.4.6
Stiffness, warp only: At standard conditions At 0 ± 2°F:	 4.4.7.1 4.4.7.2
Adhesion of coating	4.4.9
Blocking	4.4.10
Crocking, all types (dry only)	AATCC-8
Water absorption	5502 3/
Spectral reflectance	4.4.12

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TABLE IV. End item tests

Characteristic	Test Method *
Pattern execution	4.4.13
Toxicity	4.4.14
Mildew resistance	4.4.15

1/ The face side of the cloth shall be exposed to the light source. The exposure period in the weatherometer shall be 100 hours. At the end of the exposure period, the specimens shall be visually examined to determine if the coating has become stiff and brittle or soft and tacky, or if there is any evidence of cracking or crazing and then tested for hydrostatic resistance.
2/ Except that the specimen shall be stretched at 20 pounds.
3/ Except that all weighings shall be made to the nearest 0.01 gram.

* Paragraph numbers alone indicate the paragraph numbers in section 4 of this document.

4.4 Methods of inspection.

4.4.1 Hydrostatic resistance. ASTM D-751, Hydrostatic Resistance Procedure A (Pressure Application by Mullen Type Hydrostatic Tester) Procedure 1. If leakage occurs under 10 psi on any specimen, that reading shall be discarded and another specimen shall be tested. If leakage occurs under 10 psi on more than one of the five specimens, all readings shall be valid. The water pressure shall be applied only to the back side of the face cloth.

4.4.2 Resistance to high humidity. Three 4 by 4-inch specimens shall be laid flat, face side up, on a supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1 inch below the specimens. The lid of the desiccator shall be put in place and the desiccator placed in a circulating air oven having a temperature of 180°F $\pm$ 2°F for a period of 14 days. At the end of the aging period, each specimen shall be removed from the desiccator, visually examined for colorfastness and for resistance to high humidity and then tested immediately in accordance with 4.4.1.

4.4.3 Water wicking test. One 2 inch by 7 inch sample of waterproof cloth with the long dimension in the filling direction shall be the test specimen. One end of the specimen shall be weighted (10 to 20 grams) across the full width of the specimen and immersed to a depth of 2 inches in distilled water at room temperature for 5 minutes. When withdrawn from the water the height of the wetted area above the original 2-inch water line shall be immediately measured to the nearest 1/32 inch. Three determinations shall be made per sample unit.

4.4.4 Resistance to leakage test. ASTM-D-751, Hydrostatic Resistance, Procedure B, (Pressure Application by a Rising Column of Water), Procedure 2 with the hydrostatic head fixed at 50 centimeters and applied to the test specimen for 10 minutes. The back side of the finished cloth shall contact the water. The report shall only include "measurement of the appearance of water drops". Leakage of any specimen shall be considered a test failure.

4.4.5 Resistance to diethyltoluamide. The diethyltoluamide reagent for use in this test is a 75% diethyltoluamide and 25% ethanol solution (see 6.5). Three drops of the diethyltoluamide solution shall be placed in the center of a 4 by 8- inch specimen of the finished cloth with the diethyltoluamide solution contacting the face side. The specimen shall be folded to form a 4 by 4-inch square with the face sides contacting each other. The folded specimen shall then be placed between two 6 by 6-inch glass plates and a 4-pound weight placed on the assembly and left at standard conditions for 16 hours. The specimen shall then

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be removed from between the glass plates, scale rated for blocking as shown in ASTM D-751, Determination of Blocking Resistance of Fabrics Face with Rubber or Plastics at Elevated Temperatures, except that the specimen shall not be exposed to elevated temperature, and then immediately examined for conformance to the requirements in Table I.

4.4.6 Abrasion resistance . ASTM-D-3886 except that a solid rubber diaphragm 0.030 ± 0.010 inch thick with a nonmetallic contact shall be used. The abradant shall be the face side of the waterproof cloth. The face side of the waterproof cloth specimen shall be abraded 1000 multidirectional cycles. The specimen shall then be tested for hydrostatic resistance in accordance with 4.4.1 with the abraded portion of the specimen centered in the hydrostatic test area.

4.4.7 Stiffness

4.4.7.1 TAPPI Method T-451, Preferred Procedure (1) except that five test specimens with the long dimension parallel to the warp direction of the finished cloth shall be tested and that the standard textile test conditions as specified in ASTM D-1776 shall be used.

4.4.7.2 The stiffness test shall be conducted as specified in 4.4.7.1 except that the apparatus and test specimens shall be subjected to a temperature of 0°F + 2°F for a period of 4 hours and the test shall be performed in a still atmosphere at that temperature.

4.4.8 Accelerated weathering. AATCC Method 169 except that the deviations shown below shall apply. At the end of the accelerated weathering exposure period, each specimen shall be visually examined for colorfastness and for resistance to accelerated weathering and then tested for hydrostatic resistance.

Deviations to AATCC Method 169:

a. The test apparatus shall be either test chamber type 1A or IB. Type 1B shall be equipped with a three-tiered inclined specimen rack. The apparatus shall be equipped with an automatic light monitor and shall be capable of automatically controlling irradiance, temperature, and humidity. The apparatus shall be maintained in accordance with the manufacturer's recommendations.

b. The weathering test cycle shall be 40 minutes of light, 20 minutes of light with water spray on the fabric face, 60 minutes of light, 60 minutes of darkness. The test cycle shall be repeated until the total energy exposure is equal to 100 kilojoules per square meter.

c. The irradiance level shall be 0.55 ± 0.01 watt/square meter/nanometer (w/m2/nm) bandpass at 340 nanometers.

d. The glass filter combination shall be a quartz inner filter and a borosilicate type "S" outer filter.

e. The relative humidity shall be 50 ± 5 percent during the light cycle and not lower than 95 percent during the dark cycle.

f. The control set points shall be as follows:

	Dark cycle	Light cycle
Black panel	38°C	77°C
Conditioning water	40°C	53°C
Wet bulb depression 1/	0°C	10°C

1/ As a guide only; adjust to achieve required relative humidity (see e. above).

g. The test specimens shall fit smoothly on the specimen rack of the apparatus with no wrinkles or gaps. The test specimen shall be mounted to the outside of the rack with the use of appropriate stainless steel spring clips. After the required exposure period, the specimens shall be removed from the apparatus and allowed to dry and condition at Standard Conditions.

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NOTE: Monitoring of dry bulb temperature, wet bulb depression, irradiance, and black panel temperature is recommended through the use of chart recorders.

4.4.9 Adhesion of coating. ASTM-D-751, Adhesion of Coating with 2 inch wide reinforced coating adhesion specimens, cyanoacrylate (solventless) adhesive, and pulling clamp speed of 5 mm/s. Three test specimens shall be tested for each test surface, as appropriate. Test specimens shall be prepared by adhering only coating or film surfaces of test strips together in such a manner that like surfaces are adhered (i.e., face-to face or back-to-back).

4.4.10 Blocking. ASTM D-751, Determination of Blocking Resistance of Fabrics Face with Rubber or Plastics at Elevated Temperatures, except that only one specimen shall be exposed at an oven temperature of 180°F $\pm$2°F for 30 minutes.

4.4.11 Color matching. The color of the waterproof cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 $\pm$ 200°K, with illumination of 100 $\pm$ 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 $\pm$ 200°K.

4.4.12 Spectral reflectance tests. Spectral reflectance data shall be determined on the face side and shall be obtained from 600 to 860 nm respectively, at 20 nm intervals on a spectrophotometer relative to a barium sulfate standard, the preferred white standard. Other white reference materials may be used, provided they are calibrated to absolute white; e.g., magnesium oxide, or vitrolite tiles. The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. The specimen shall be measured as a single layer backed with two layers of the same fabric and shade. Measurements shall be taken on a minimum of two different areas, and the data averaged. The measurement areas should be at least 6 inches away from the edges of the finished cloth. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified in 3.5 shall be considered a test failure.

4.4.13 Pattern execution. The pattern on the cloth shall be matched to the pattern drawing (see 6.4).

4.4.14 Toxicity assessment. The cloth shall be tested for toxicity as follows:

a) Title 40, Code of Federal Regulations, 1994 Edition;
   Section 798.4100- Dermal Sensitization
   Section 798.4470- Primary Dermal Irritation
   Section 798.4500- Primary Eye Irritation
   Marzulli, F. & H. Maibach, "Contact Allergy: Predictive Testing in Humans", Advances in Modern Toxicology, Volume 4, pp 353-372, 1977.
b) As an alternative to animal and human testing, the contractor may provide information which certifies that the finished product is composed of chemicals or materials which have been safely used commercially where prolonged skin contact has occurred.

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4.4.15 Mildew resistance. A certificate of compliance shall be submitted for this requirement.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the fabrication of ground cloths and components of jungle hammocks.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
d. When first article is required (see 3.1, 4.2, and 6.3).
e. Width required.
f. Levels of preservation and packing (see 5.1).

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample and pattern drawing. For access to the shade sample (see 3.1) and pattern drawings (see 3.6), address the contracting activity issuing the invitation for bids.

6.5 Testing reagents and materials.

6.5.1 DEET Insect Repellent. The insect repellent reagent will be a solution of 75% by weight (min) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution will be a technical grade and contain N, N-diethyl-m-toluamide of not less than 95% purity and the remainderwill consist of entirely or a mixture of ortho or paraisomers of N, N-diethytoluamide. The denatured alcohol component of the solution wi ll be ethanol, U.S.P. 94.9% by volume and denatured in accordance with The Code of Federal

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Regulations 27 CFR 21, Formula 40 (see 2.1). The insect repellent must be registered with the U.S. Environmental Protection Agency in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (see 2.1).

6.6 Note. The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.

6.7 Subject term (key word) listing.
Camouflage
Parka
Trousers
Wet Weather clothing system
Woodland

Custodians:
Army - GL
Navy - NU
Air Force - 99

Review activities:
Navy - MC
Air Force 45, 82

Preparing activity
DLA-CT

Project Number
8305-0633

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-43473G	**2. DOCUMENT DATE *(YYMMDD)*** 97 December 02

3. DOCUMENT TITLE

CLOTH, WATERPROOF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER DPSC-FNS	b. TELEPHONE *(Include Area Code)* (1) Commercial (215) 737-8079 (2) AUTOVON 444-8079
c. ADDRESS *(Include Zip Code)* 2800 South 20th STREET PHILADELPHIA, PA 19145-5099	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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